

Boundedness of varieties of general type

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Outline of the talk

1 Introduction

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2 Curves

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Polynomial equations

Algebraic Geometry is concerned with the study of solutions of polynomial equations; say n -equations in m -variables

$$\begin{cases} P_1(x_1, \dots, x_m) = 0 \\ \dots \\ P_n(x_1, \dots, x_m) = 0. \end{cases}$$

Throughout this talk I will always look for complex solutions $(z_1, \dots, z_m) \in \mathbb{C}^m$.

Projective space

- It is convenient to work on a compactification $\mathbb{C}^N \subset \mathbb{P}_{\mathbb{C}}^N$ where $\mathbb{P}_{\mathbb{C}}^N = \{\mathbb{C}^{N+1} - (0, \dots, 0)\} / \{\mathbb{C}^*\}$ is projective N -space.

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- We have that $\mathbb{P}_{\mathbb{C}}^0$ is a point, $\mathbb{P}_{\mathbb{C}}^1 \cong S^2$ is the Riemann sphere and $\mathbb{P}_{\mathbb{C}}^N = \mathbb{C}^N \cup \mathbb{P}_{\mathbb{C}}^{N-1}$.

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- We then replace $P(x_1, \dots, x_N)$ by the corresponding homogeneous polynomial in $\mathbb{C}[x_0, \dots, x_N]$. Eg.
 $x_1^2 + x_2^3 \Rightarrow x_0 x_1^2 + x_2^3$.

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- We then have:

Bezout's Theorem

Theorem (Bezout's Theorem)

Let $P_1(x_0, \dots, x_n), \dots, P_n(x_0, \dots, x_n)$ be n homogeneous polynomials of degrees $d_1, \dots, d_n$. If the set

$$\bigcap_{i=1}^n \{P_i(x_0, \dots, x_n) = 0\} \subset \mathbb{P}_{\mathbb{C}}^n$$

is finite, then it consist of exactly $d_1 \cdots d_n$ points (counted with multiplicity).

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- For most choices of polynomial equations, the solution set is smooth.
- By Hironaka's Theorem, it is possible to “desingularize” singular varieties.

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- The simplest curves are **rational curves** i.e. curves of genus 0.
- There is only one such curve $X = \mathbb{P}_{\mathbb{C}}^1$.
- When $g = 1$ we have a 1-parameter family of **elliptic curves** $X_t = \{y^2 - x(x - 1)(x - t) = 0\}$.

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$$X = \bigcap_{i=1}^n \{P_i(x_1, \dots, x_m) = 0\} \subset \mathbb{P}_{\mathbb{C}}^m.$$

- We would like to find a natural (canonical) description for $X \subset \mathbb{P}_{\mathbb{C}}^m$.

Line bundles

- Since X is compact, the only global holomorphic functions are constant. Therefore, we consider meromorphic functions with prescribed poles or equivalently sections of line bundles.

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- If L is a line bundle on X and $s_0, \dots, s_m$ is a basis of $H^0(X, L)$, then there is a map $\phi_L : X \dashrightarrow \mathbb{P}_{\mathbb{C}}^m$ given by

$$x \rightarrow [s_0(x) : \dots : s_m(x)]$$

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- To define an embedding of X in $\mathbb{P}_{\mathbb{C}}^m$, we must find a line bundle L whose sections distinguish different points and tangent directions.

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Theorem

Let X be a curve of genus $g \geq 2$, then the sections of $H^0(X, \omega_X^{\otimes 3})$ define an embedding of degree $6g - 6$

$$X \hookrightarrow \mathbb{P}_{\mathbb{C}}^{5g-6}.$$

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- To obtain a compact moduli space, we must allow stable curves (nodal singularities).

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- If $g \geq 2$, then $\omega_X^{\otimes 3} = \mathcal{O}_{\mathbb{P}^N}(1)|_X$. It follows that $R(X)$ is finitely generated $\mathbb{C}$ -algebra which captures the geometry of X . In fact $X = \text{Proj} R(\omega_X)$.

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- In coordinates this may be described as follows. If $r_0, \dots, r_m$ are generators of $R(\omega_X)$ and $K = \ker(\mathbb{C}[x_0, \dots, x_m] \rightarrow R(\omega_X))$, then $X \subset \mathbb{P}_{\mathbb{C}}^m$ is defined by the equations in K .

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- The **Kodaira dimension** of X is $\kappa(X) = \text{tr.deg.}_{\mathbb{C}} R(\omega_X) - 1$.
- We have $\kappa(X) = \max_{k \geq 0} \{\dim \phi_{\omega_X^{\otimes k}}(X)\}$ so that $\kappa(X) \in \{-1, 0, 1, \dots, d\}$.

Birational geometry of surfaces

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- Consider for example a smooth surface $X \subset \mathbb{P}^3$ given by a general degree 5 homogeneous polynomial.
- By adjunction $\omega_X = \mathcal{O}_X(1)$ and $X \cong \text{Proj} R(\omega_X)$.
- Given any point $x \in X$, consider now the **blow up** of X at x :
 $\nu : \tilde{X} = \text{Bl}_x(X) \rightarrow X$.

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- The corresponding morphism is just $\tilde{X} \rightarrow X \cong \operatorname{Proj} R(\omega_{\tilde{X}})$.

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- It turns out that two surfaces X and X' are birational if and only if there is a sequence of blow ups
$$X_n \rightarrow X_{n-1} \rightarrow \dots X_1 \rightarrow X \text{ and } X'_m \rightarrow X'_{m-1} \rightarrow \dots X'_1 \rightarrow X'$$
such that $X_n = X'_m$.

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- It turns out that two surfaces X and X' are birational if and only if there is a sequence of blow ups $X_n \rightarrow X_{n-1} \rightarrow \dots X_1 \rightarrow X$ and $X'_m \rightarrow X'_{m-1} \rightarrow \dots X'_1 \rightarrow X'$ such that $X_n = X'_m$.
- The canonical ring is a birational invariant.

Birational geometry of surfaces IV

- By a Theorem of Castelnuovo, one may reverse the blowing up procedure i.e. given any surface X and any -1 curve $E \subset X$, there exists a morphism $\nu : X \rightarrow X_1$ such that $X = \text{Bl}_x(X_1)$ for some point $x \in X_1$.

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- Note that $\rho(X_1) = \rho(X) - 1 \geq 0$ and so this procedure can be repeated at most finitely many times.
- A **minimal surface** is a surface that contains no -1 curves.
- By Castelnuovo's Theorem, given any surface X there exists a birational morphism to a minimal surface $X \rightarrow X_{\min}$ (given by finitely many blow downs).

Birational geometry of surfaces V

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- If $\kappa(X) = 0$, then $X_{\min}$ is a K3, abelian, Enriques or hyperelliptic surface.
- If $\kappa(X) = 1$, then there is a morphism $X_{\min} \rightarrow B$ whose general fiber F is a curve of genus 1; $\omega_{X_{\min}} \cdot F = 0$.

Birational geometry of surfaces VI

- If $\kappa(X) = 2$, then by a result of Bombieri, $\phi_{\omega_{X_{min}}^{\otimes 5}} : X_{min} \rightarrow \mathbb{P}^N$ is a morphism whose image $X_{can} \cong \text{Proj } \bigoplus_{m \geq 0} H^0(\omega_X^{\otimes m})$ is given by contracting curves $\mathbb{P}^1 \cong E \subset X_{min}$ with $\omega_{X_{min}} \cdot E = 0$.

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- X_{min} is the minimal desingularization of X_{can} .
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- X_{min} is the minimal desingularization of X_{can} .
- X_{can} has Du Val singularities and degree $25K_{X_{min}}^2$ in $\mathbb{P}^N$. (K_X is given by the zeroes of a meromorphic section of ω_X .)
- Therefore, if $K_{X_{min}}^2 \leq M$ then there are only finitely many families of such surfaces (canonical/minimal).

Birational geometry of surfaces VII

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- Note that the moduli space of K3 surfaces has countably many irreducible components! (The problem is that $\omega_X = \mathcal{O}_X$ so one can not use multiples of the canonical line bundles to study X .)
- The moduli spaces of (canonically polarized) surfaces of general type were constructed by Gieseker.

Outline of the talk

- 1 Introduction
- 2 Curves
- 3 Surfaces
- 4 Higher dimensions.

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- They showed that if X is not covered by $\mathbb{P}^1$'s, then there exists a finite set of well understood birational maps (known as flips and divisorial contractions) $X \dashrightarrow X_1 \dashrightarrow \dots \dashrightarrow X_N$ such that ω_{X_N} is semiample (i.e. $\phi_{\omega_{X_N}^{\otimes m}} : X_N \rightarrow \mathbb{P}^N$ some $m > 0$).

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- In particular, $R(\omega_X)$ is finitely generated.

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- Flips $X \dashrightarrow X^+$ are a new operation that only exists in dimension ≥ 3 .
- K_X -negative curves are replaced by K_{X^+} -positive curves.
- The main issue is to prove the existence of flips and to show that there are no infinite sequences of flips.

Higher dimensional MMP

Recently there have been several important developments in higher dimensions ($d \geq 4$). In particular we have:

Theorem (Birkar-Cascini-Hacon-M^cKernan (2006))

Let X be a variety of general type (i.e. $\kappa(X) = \dim(X)$), Then there exists a finite sequence of flips and divisorial contractions

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- $X_N = X_{min}$ is a minimal model.
- Flips exist (in all dimensions).

Finite generation

We can also show that:

Theorem (Birkar-Cascini-Hacon-M^cKernan, Siu (2006))

The canonical ring

$$R(\omega_X) = \bigoplus_{k \geq 0} H^0(X, \omega_X^{\otimes k})$$

is finitely generated.

$X_{\text{can}} = \text{Proj} R(\omega_X)$ is the canonical model.

Boundedness

One can also generalize some of the other features of the classification of surfaces. For example:

Theorem (Tsuji, Hacon-M^cKernan, Takayama)

For any integers $d, M > 0$, there are only finitely many families of varieties of general type, dimension d and volume $\text{Vol}(\omega_X) = K_{X_{\min}}^d \leq M$ modulo birational isomorphisms.

Moduli spaces

Combining this with the existence of canonical models, one obtains

Theorem

For any integers $d, M > 0$, there are exist a morphism $f : \mathcal{X} \rightarrow B$ (of normal quasi-projective varieties) whose fibers parametrize varieties X with mild (canonical) singularities such that ω_X is ample and $K_X^d \leq M$.

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In fact there exists a moduli space for canonically polarized varieties of dimension d and fixed K_X^d .

Consequences

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- However, if $d = 3$, using Reid's exact plurigenera formula, J.-A. Chen and M. Chen show that $r_0 \leq 77$ and $V \geq 1/2660$.
- There are examples (Iano-Fletcher) showing that if $d = 3$, $r_0 \geq 27$ and $V \leq 1/420$.

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- To prove $r_0 \leq A + B/(K_X^d)^{1/d}$ we proceed as follows.

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- Therefore we have a section of $H^0(\omega_X^{\otimes \lambda+1})$ which doesn't vanish at x .

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- In **very rough terms**, the idea is then that by Kawamata sub-adjunction $\omega_X^{\otimes 1+\lambda}|_V \geq \omega_V$.
- Using the MMP, we can assume ω_X is ample and so by Serre Vanishing we can lift these sections to get the required sections of $\omega_X^{\otimes \rho(1+\lambda)}$.

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- Once again the key issue is to show that these varieties (or better the pairs (X_i, D_i)) are bounded.
- We have the following (work in progress!).

DCC for volumes

Theorem (Hacon, McKernan, Xu)

Let

$V = \{(K_X + D)^d \mid (X, D) \text{ is lc, } K_X + D \text{ is ample and } D \text{ is reduced}\}.$
Then V satisfies the DCC. In particular there is a minimal positive element $0 < v \in V$.

One can use this to deduce results about the existence of moduli spaces of canonically polarized stable (semi-log-canonical) varieties. Note that V does not satisfy the ACC (even if $D = 0$).

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Theorem (Hacon, McKernan, Xu)

The ACC for lct holds in $\dim \leq 4$ and in dimension ≥ 5 there are no rational accumulation points from below.

By Ein-deFernex-Mustata, the ACC for lct is known for all smooth varieties.

Conclusion

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- **Apologies** to the many mathematicians whose work I should have also mentioned in more detail. Amongst these Alexeev, Kawamata, Kollár, Mori, Reid, Shepherd-Barron, Siu, Shokurov, Viehweg and others!